

Circadian rhythms regulate nearly every physiological process in mammals. The suprachiasmatic nucleus (SCN) in the hypothalamus acts as the master clock, synchronising peripheral oscillators in every organ. Light is the primary zeitgeber: retinal ganglion cells project directly to the SCN via the retinohypothalamic tract. Disruption of circadian timing - through shift work, jet lag, or chronic sleep restriction - elevates cortisol, impairs glucose metabolism, and is associated with increased cancer risk. Melatonin, secreted by the pineal gland in the absence of light, signals night onset and has both chronobiotic and antioxidant properties at physiological concentrations. Chronotherapy exploits predictable oscillations in drug metabolism to time dosing for maximum efficacy and minimum side effects. Temperature is an underappreciated zeitgeber: a 1-2 degree drop in core body temperature initiates sleep onset. Modern smartphones emit blue light (400-490 nm) that suppresses melatonin production. Recovery sleep after acute deprivation restores cognitive performance but may not fully reverse immune or metabolic consequences of chronic restriction. [section 0]

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